

PUTGARDEN KAP ARKONA, Germany

WMO: 100910

Lat: 54.6792N

Lon: 13.4342E

Elev: 137

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.4	21.4	10.6	9.5	19.0	14.4	11.5	22.8	50.8	35.4	45.8	37.3	15.9	0	0.841

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.1	73.5	64.7	71.1	63.8	68.9	62.5	66.3	72.0	64.7	69.8	63.3	67.9	11.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.9	89.5	69.4	62.4	84.8	68.0	61.0	80.5	66.6	31.0	72.2	29.8	69.9	28.7	68.1	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
42.7	38.0	34.0	16.6	80.5	8.3	3.7	10.7	83.2	5.8	85.3	1.1	87.4	-4.9	90.0
			15.1	69.8	7.7	2.0	9.6	71.3	5.1	72.5	0.7	73.6	-4.9	75.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	34.0	33.2	37.3	42.3	50.6	57.8	62.5	62.5	56.7	50.1	41.5	36.7	(6)
	DBStd	11.66	6.80	6.62	5.58	4.92	4.98	4.45	4.07	4.10	3.77	5.10	4.57	4.83	(7)
	HDD50	2380	495	471	394	237	54	1	0	0	1	61	254	412	(8)
	HDD65	6552	960	891	858	680	447	221	99	101	251	463	704	877	(9)
	CDD50	1359	0	0	1	6	73	235	389	389	202	63	1	0	(10)
	CDD65	54	0	0	0	0	0	4	23	25	2	0	0	0	(11)
	CDH74	76	0	0	0	0	0	6	23	46	1	0	0	0	(12)
	CDH80	5	0	0	0	0	0	0	1	4	0	0	0	0	(13)
(14) Wind	WSAvg	17.0	21.1	19.0	17.7	16.0	14.4	13.9	14.6	14.5	16.8	17.6	18.5	19.8	(14)
(15) Precipitation	PrecAvg	20.8	1.3	1.1	1.3	1.5	1.7	1.6	2.2	2.4	2.0	1.8	2.2	1.7	(15)
	PrecMax	29.3	2.1	2.0	3.0	3.7	3.2	3.0	4.1	5.7	4.1	4.7	5.0	3.4	(16)
	PrecMin	13.1	0.4	0.2	0.4	0.1	0.5	0.5	0.4	0.4	0.5	0.2	0.7	0.6	(17)
	PrecStd	4.1	0.6	0.6	0.7	0.9	0.7	0.7	1.3	1.5	0.9	1.3	1.2	0.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.8	53.3	55.0	60.3	66.6	73.4	76.6	79.3	71.3	64.6	53.1	49.8	(19)
		MCWB	46.6	49.4	49.3	52.2	57.7	64.4	66.5	66.7	63.6	59.9	49.9	47.1	(20)
	2%	DB	45.1	47.4	49.2	55.4	62.4	69.9	73.5	74.0	66.8	60.2	50.6	47.7	(21)
		MCWB	43.1	44.1	45.8	48.1	55.7	63.0	65.0	65.1	61.1	56.3	48.3	45.8	(22)
	5%	DB	43.7	43.4	46.1	51.8	60.1	67.0	71.2	70.9	63.9	58.0	49.1	45.0	(23)
		MCWB	41.7	40.8	43.2	46.8	54.3	60.6	63.9	64.1	59.4	54.5	47.0	43.3	(24)
	10%	DB	42.2	41.0	44.0	49.0	58.0	64.6	68.8	68.7	62.0	56.3	47.6	43.4	(25)
		MCWB	40.4	38.9	41.5	45.1	52.9	58.9	62.5	62.7	57.7	53.0	45.6	41.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.7	49.5	50.2	52.9	59.1	66.8	68.2	68.8	64.9	60.2	50.4	48.1	(27)
		MCDB	48.1	52.4	53.7	59.9	65.2	71.1	74.1	75.3	69.8	64.7	51.8	49.2	(28)
	2%	WB	43.6	43.9	46.1	49.2	56.7	63.6	66.1	66.6	62.1	56.8	48.6	45.7	(29)
		MCDB	44.9	46.8	48.7	53.8	61.4	68.3	71.9	72.4	65.5	59.4	50.1	47.4	(30)
	5%	WB	42.2	41.1	43.7	47.4	54.9	61.5	64.6	64.9	60.1	55.0	47.3	43.7	(31)
		MCDB	43.5	43.1	45.8	51.2	59.4	66.2	69.9	69.9	63.2	57.4	48.8	45.1	(32)
	10%	WB	40.5	39.1	41.6	45.4	53.3	59.4	63.2	63.4	58.2	53.6	45.8	41.9	(33)
		MCDB	42.0	40.9	43.7	48.6	57.3	64.1	68.1	67.7	61.6	55.8	47.4	43.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	5.9	6.7	8.3	8.7	8.8	8.8	9.1	7.6	6.7	5.9	6.0	(35)
		MCDBR	6.0	8.7	10.7	13.9	11.7	11.9	12.0	12.0	10.7	8.4	6.8	7.2	(36)
	5% WB	MCWBR	5.9	7.1	8.2	9.0	8.2	7.9	7.7	7.1	7.6	6.9	6.5	6.9	(37)
		MCWBR	5.9	8.5	10.3	12.5	10.8	11.4	11.0	10.8	10.1	7.7	6.6	6.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.319	0.341	0.367	0.396	0.392	0.387	0.408	0.409	0.377	0.362	0.341	0.311	(40)
		taud	2.327	2.337	2.300	2.298	2.370	2.411	2.368	2.374	2.442	2.419	2.364	2.310	(41)
	Ebn at Noon		196	234	254	262	271	273	264	256	250	224	185	171	(42)
		Edh at Noon	18	25	32	36	36	35	36	33	28	23	17	15	(43)
(44) All-Sky Solar Radiation	RadAvg		151	344	757	1321	1691	1814	1676	1357	969	493	197	105	(44)
	RadStd		21	62	69	150	178	132	193	96	85	69	26	11	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (55 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-286	+108	N/A

Nomenclature: See separate page